Code No. 3501/S

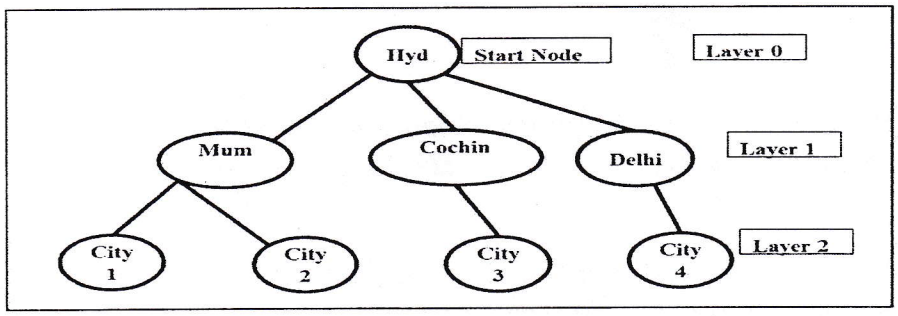
OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (**CSE**) V-Semester **(Suppl.)** Examinations **July/August 2025**

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

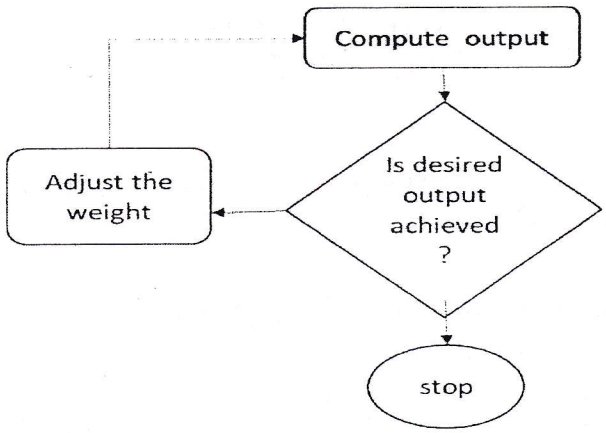
Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7), each question carries **12** Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any, may suitably be assumed.***

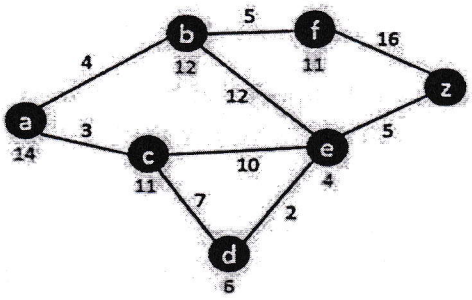
		Marks	BT	CO
1 a)	Given the graph of cities below, if you apply DFS and BFS algorithms to reach City4, what search path you will follow? Mention the path in both cases.	2	2	1



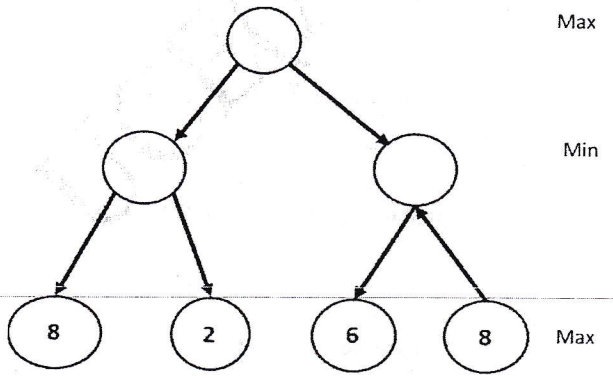
b)	What are the different types of agents in knowledge representation? Give example.	2	1	1
c)	Explain the four (4) REs in CBR (Case Based Reasoning) in Expert Systems with flow chart.	2	3	2
d)	Following is the typical ANN learning flowchart . What do you understand by the step" Adjust weight".	2	4	2



e)	What are the semantics of propositional logic, and how do they determine meaning?	2	2	2
f)	What is the use of KNN? When is it most suitable to be used?	2	5	3
2 a)	What is BFS algorithm and its application in AI Informed search mechanism? Write down the algorithm.	5	3	1
b)	Apply the A* search to find the solution path from a to z. Heuristics are with nodes, and cost is with edges. Write all steps as well as open and closed lists for full marks.	7	6	3



3 a)	Explain Alpha-Beta Pruning in Minimax Algorithm. Use a suitable example with a game tree to demonstrate the process of pruning.	6	2	2
b)	Use the Alpha-Beta pruning algorithm to prune the game tree in Problem 1 (a) assuming child nodes are visited from left to right. Which nodes are cut off? Circle the nodes that are not examined and label them with X.	6	5	3



4 a)	What are the components (architecture) in a typical expert system? Explain with neat diagram.	6	1	2
b)	What is uncertainty in Artificial Intelligence? Explain the primary sources of uncertainty in AI systems.	6	2	1

5	a)	In machine learning explain what is ID3 learning algorithm and how it decides the root attribute using entropy?	5	4	3
	b)	For deciding the play, following the play dataset contains the following attributes:	7	6	3

- Weather: {Sunny, Overcast, Rainy}
- Temperature: {Hot, Mild, Cool}
- Play: {Yes, No}

Weather	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Cool	Yes
Rainy	Cool	Yes
Rainy	Mild	Yes
Rainy	Hot	No

i) Calculate the Entropy of the dataset (Play)
ii) Calculate the Entropy of the weather
iii) Calculate the Entropy of the temperature
iv) Calculate the infogain of weather

6	a)	What is natural language processing? What are the process steps in natural language processing?	6	2	1
	b)	Briefly explain what are the statistical methods used in NLP.	6	1	2
7	a)	Provide the steps in K means algorithm and purpose and how we use this in AI/ML?	5	2	3
	b)	A well-known health organization has carried out medical diagnostic test and collected following data for detecting a rare disease. Out of 1,000 patients, 100 actually have the disease. The test's confusion matrix is as follows:	7	5	3

	Predicted Positive	Predicted Negative
Actual Positive	90	20
Actual Negative	100	610

i) Calculate the accuracy of the test.
ii) Calculate recall and precision, and explain their significance in this context

******